Techniques of Integration: Partial Fractions, Parts, and Substitution

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§1.1 Integration methods are reversed differentiation rules

Before selecting a method, it helps to ask what derivative could have produced the integrand. Three common sources are the chain rule, product rule, and differentiation of logarithms or inverse trigonometric functions. Their inverses are substitution, integration by parts, and integration of partial fractions. The elementary organization here follows *OpenStax, Calculus Volume 2*, Chapter 3 (Integration Techniques).[1]

Linearity is the permission to deal with the resulting pieces independently. On any interval on which the functions have antiderivatives,

$$\int (f + g - h)\,dx = \int f\,dx + \int g\,dx - \int h\,dx.$$

All equalities between *indefinite* integrals are understood up to an additive constant on each connected interval of definition. This qualification matters when a denominator vanishes: crossing a pole need not preserve the same choice of constant.

§1.2 Partial fractions begin with polynomial algebra

The first example is

$$I = \int \frac{dx}{x^2 - a^2}, \qquad a \neq 0.$$

The factorization, rather than any rule of integration, determines the answer:

$$\frac{1}{(x-a)(x+a)} = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right).$$

Consequently, on an interval avoiding the two poles,

$$\boxed{I = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.}$$

The signs in this formula can be checked by differentiating the logarithm, which returns $1/(x^2 - a^2)$. The factor $1/(2a)$ is not a memorized integration constant: it comes from solving a polynomial identity.

In general, for a rational function $P(x)/Q(x)$, first perform polynomial division if $\deg P \geq \deg Q$. Factor Q over the field in which the decomposition is sought. A real quadratic irreducible factor requires a linear numerator; a repeated linear factor $(x-a)^m$ requires *every* term

$$\frac{c_1}{x-a} + \frac{c_2}{(x-a)^2} + \cdots + \frac{c_m}{(x-a)^m}.$$

This is forced by the algebra of rational functions, independently of the later integration.

[1] https://openstax.org/books/calculus-volume-2/pages/3-introduction

Example 1.2.1 (A repeated pole really needs two coefficients)
Consider the original problem

$$I = \int \frac{3x+5}{x^3 - x^2 - x + 1}\, dx.$$

The denominator factors as

$$x^3 - x^2 - x + 1 = (x+1)(x-1)^2.$$

The correct ansatz is

$$\frac{3x+5}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

Multiplying through gives

$$3x + 5 = A(x-1)^2 + B(x+1)(x-1) + C(x+1).$$

Putting $x = -1$ and $x = 1$ yields $A = \frac{1}{2}$ and $C = 4$; comparing the remaining coefficient yields $B = -\frac{1}{2}$. Thus

$$\boxed{I = \tfrac{1}{2}\ln|x+1| - \tfrac{1}{2}\ln|x-1| - \frac{4}{x-1} + C.}$$

Differentiating the three summands is a direct independent check. Omitting the square-pole term would make the polynomial identity unsolvable.

For an irreducible quadratic, completing the square is the other algebraic step. Write

$$t = x + \frac{p}{2}, \qquad x^2 + px + q = t^2 + \Delta, \qquad \Delta = q - \frac{p^2}{4},$$

and $mx + n = mt + (n - \frac{mp}{2})$. The term with numerator t is a logarithm; the constant numerator produces an arctangent if $\Delta > 0$, while $\Delta < 0$ reduces to real linear factors. For $\Delta > 0$,

$$\int \frac{mx+n}{x^2+px+q}\, dx = \frac{m}{2}\ln(t^2+\Delta) + \frac{n - mp/2}{\sqrt{\Delta}} \arctan\frac{t}{\sqrt{\Delta}} + C.$$

This also identifies precisely which denominator sign produces which type of antiderivative.

§1.3 Substitution is the chain rule in reverse

Suppose $x = \rho(t)$ is differentiable. The chain rule says that if $F' = f$, then

$$\frac{d}{dt}F(\rho(t)) = f(\rho(t))\rho'(t).$$

Hence

$$\int f(\rho(t))\rho'(t)\, dt = F(\rho(t)) + C.$$

For a definite integral, the same argument gives

$$\int_{\rho(a)}^{\rho(b)} f(x)\,dx = \int_a^b f(\rho(t))\rho'(t)\,dt$$

under the usual regularity hypotheses; the signed derivative is intentional.

For example, the inner derivative in

$$\int \sin^3 x \cos x\,dx$$

is visible. Taking $t = \sin x$ gives

$$\int t^3\,dt = \frac{t^4}{4} + C = \boxed{\frac{\sin^4 x}{4} + C}.$$

By contrast, $\int \sin^2 x\,dx$ is easier after a trigonometric identity:

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The preliminary algebraic transformation is as important as the substitution.

Example 1.3.1 (A square root becomes a trigonometric identity)
For $a > 0$ and $|x| < a$, put $x = a\sin t$ with $t \in (-\pi/2, \pi/2)$. Then

$$\sqrt{a^2 - x^2} = a\cos t, \qquad dx = a\cos t\,dt,$$

so

$$\int \sqrt{a^2 - x^2}\,dx = a^2 \int \cos^2 t\,dt = \frac{a^2}{2}\,(t + \sin t\cos t) + C.$$

Returning to x gives

$$\boxed{\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C.}$$

One obtains related substitutions

$$\sqrt{a^2 + x^2} : x = a\tan t, \qquad \sqrt{x^2 - a^2} : x = a\sec t$$

on intervals where the signs and inverse branches are chosen consistently.

§1.4 Integration by parts is the product rule in reverse

For differentiable u, v, integrate $(uv)' = u'v + uv'$ to obtain

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

For $\int \ln x\,dx$ on $x > 0$, choosing $u = \ln x$ and $dv = dx$ turns the logarithm into $1/x$:

$$\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C.$$

The useful choice is the one that makes the *new* integral simpler, not necessarily the one that makes one factor shorter.

Repeated integration by parts yields a finite identity. If u has $n+1$ derivatives and $v^{(n+1)}$ is the other factor, then

$$\int u\, v^{(n+1)}\, dx = uv^{(n)} - u'v^{(n-1)} + \cdots + (-1)^n u^{(n)} v + (-1)^{n+1} \int u^{(n+1)} v\, dx.$$

For a polynomial $u = P$ of degree n, the last integral vanishes. Thus

$$\int P(x) e^x\, dx = e^x (P - P' + P'' - \cdots + (-1)^n P^{(n)}) + C.$$

In particular, for $P = 2x^3 + 3x^2 + 4x + 5$,

$$P' = 6x^2 + 6x + 4, \quad P'' = 12x + 6, \quad P''' = 12,$$

and

$$\boxed{\int (2x^3 + 3x^2 + 4x + 5) e^x\, dx = e^x (2x^3 - 3x^2 + 10x - 5) + C.}$$

Differentiate this answer once to check every coefficient. Repeated parts with $P(x)\sin x$ or $P(x)\cos x$ works similarly, but the successive antiderivatives cycle through sines and cosines.

§1.5 The same calculus rules survive in differential forms

Substitution is the one-dimensional instance of the pullback of a 1-form:

$$\rho^*(f(x)\, dx) = f(\rho(t))\rho'(t)\, dt.$$

This language records the derivative factor automatically. The higher-dimensional change-of-variables formula has an absolute Jacobian determinant for ordinary area,

$$\iint_{\rho(U)} f(x,y)\, dx\, dy = \iint_U f(\rho(u,v))\, |\det D\rho(u,v)|\, du\, dv$$

when ρ is a suitable one-to-one regular change of coordinates. Oriented differential forms use the *signed* determinant instead. The distinction avoids confusing substitution with an arbitrary change of labels.

The definite form of integration by parts also keeps track of boundary information:

$$\int_a^b (Du) v\, dx = [uv]_a^b - \int_a^b u(Dv)\, dx, \qquad D = \frac{d}{dx}.$$

If the boundary term vanishes on a chosen domain of functions, D is formally skew-adjoint with respect to the real L^2 pairing. Without boundary conditions it would be misleading to write simply $D^* = -D$. This is the first elementary example of a boundary term controlling an adjoint differential operator.

Finally, partial fractions have their complex-analytic interpretation: the coefficient of $(z-a)^{-1}$ in the Laurent principal part is the residue. When f is meromorphic inside a positively oriented closed contour γ and has no poles on γ,

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{a \text{ inside } \gamma} \operatorname{Res}(f, a).$$

This gives the precise link from the algebra of rational denominators to complex integration. The three elementary methods have different mechanisms—factorization, the product rule, and the chain rule—and these mechanisms remain identifiable in more advanced mathematics.